ME-205

DESIGN LAB-1

Presented by: TU-E

Instructor:

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TITLE - Experimental Determination of Young's Modulus and Poisson's Ratio Using Strain Gauges and Wheatstone Bridge

OBJECTIVE - The objective of this project is to determine the mechanical properties of a material, namely Young's Modulus and Poisson's Ratio, by measuring strain using strain gauges. A Wheatstone bridge circuit is used to detect small changes in resistance caused by deformation, and an amplifier is employed to enhance the signal for accurate measurement. The recorded data is then used to establish the stress–strain relationship and evaluate the material properties with precision.

Click the link for seeing the - Video Demonstration

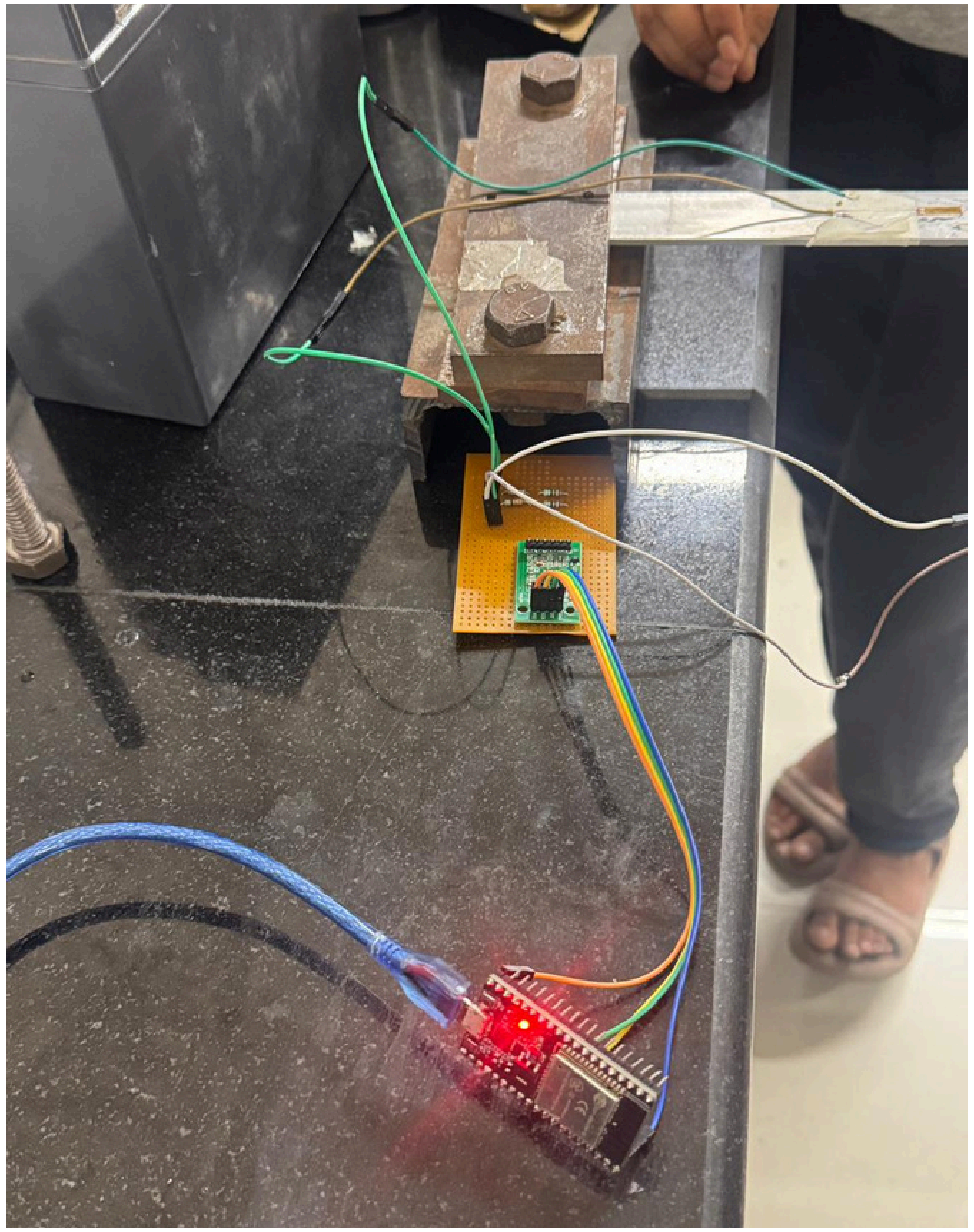
APPARATUS::

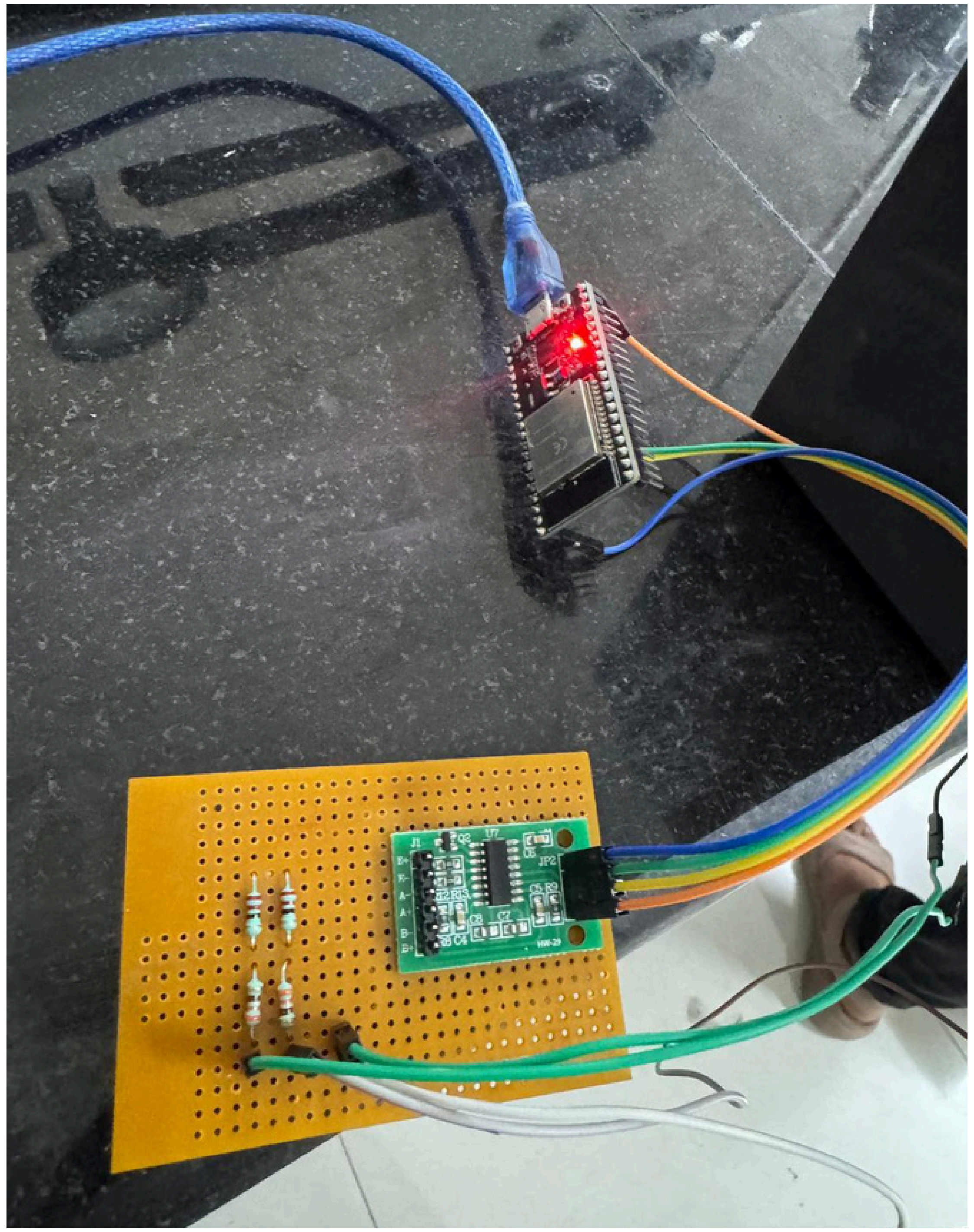
The following components were used to set up the experimental system for measuring strain and determining material properties:

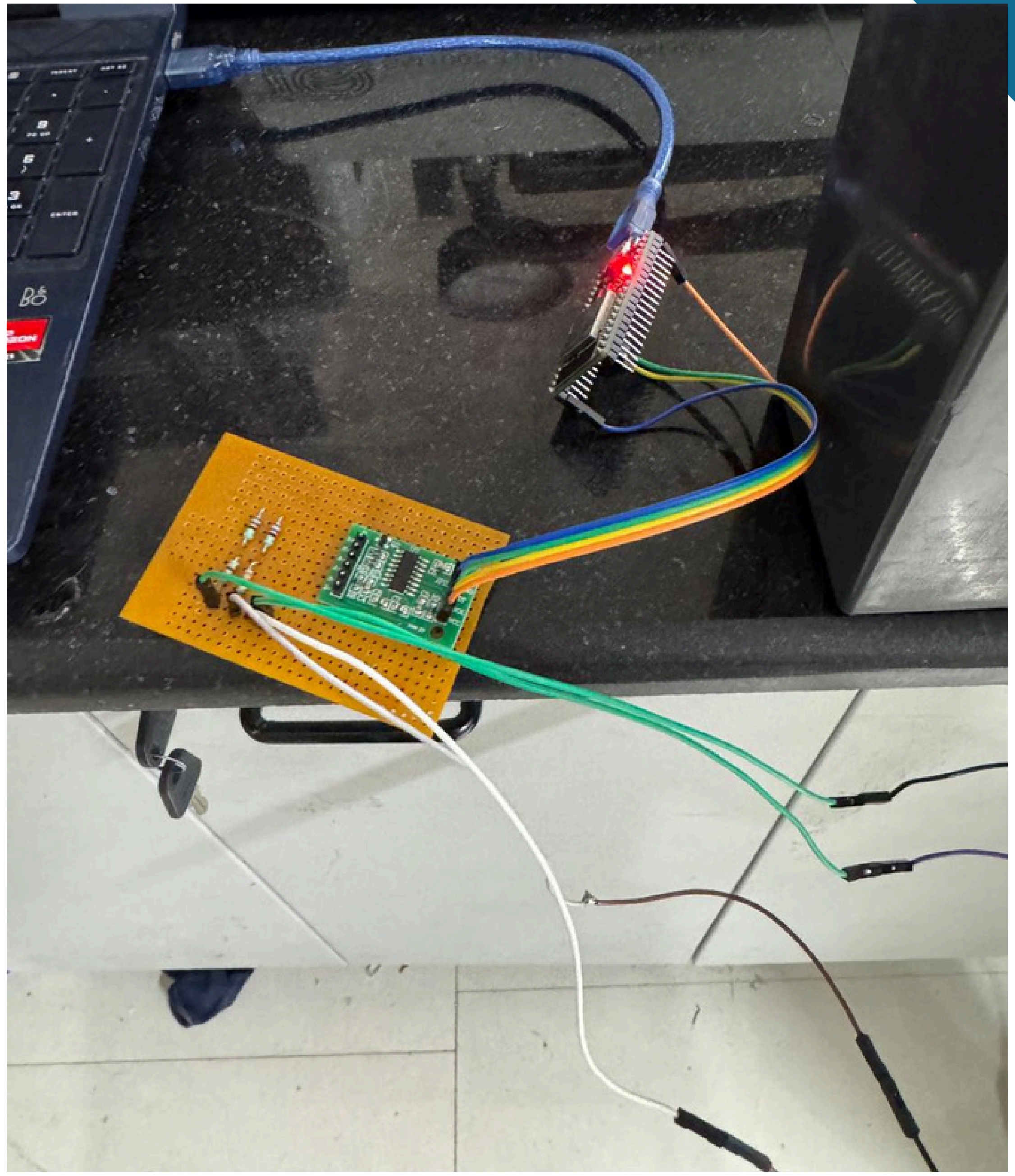
- Strain Gauges (350 Ω): Used to measure the deformation (strain) in the material by converting mechanical changes into electrical resistance variations.
- Resistors (350 Ω, 3.3 kΩ, 1.8 kΩ, 70 Ω): Used for constructing the Wheatstone bridge and for signal conditioning within the circuit.
- HX711 Load Cell Amplifier: A high-precision amplifier used to amplify the small electrical signals obtained from the strain gauge bridge.
- ESP32 Microcontroller: Used for data acquisition, processing, and interfacing with the amplifier module.
- Potentiometer: Employed for balancing the Wheatstone bridge and fine-tuning the circuit.
- PCB (Printed Circuit Board): Used to mount and securely connect all electronic components.
- Jumper Wires: Used for making electrical connections between components during prototyping.
- Clamper (Clamp): Used to hold the specimen firmly in place during the experiment.
- Strain Gauge Adhesive Setup: Ensures proper attachment of strain gauges to the material surface for accurate measurement.

These components together form the complete setup required to measure strain accurately, process the signal, and determine Young’s Modulus and Poisson’s Ratio.

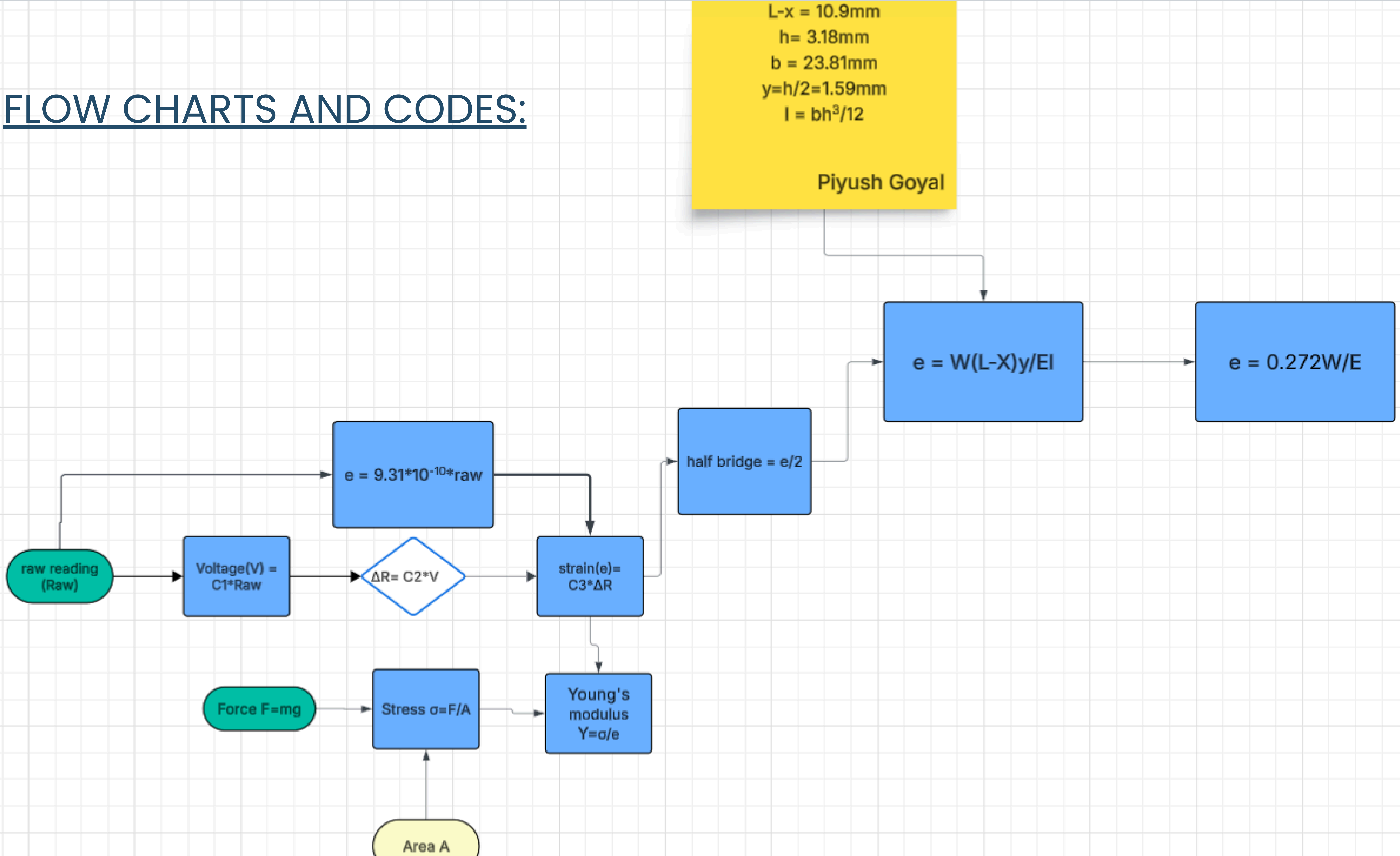
CIRCUIT::







FLOW CHARTS AND CODES:



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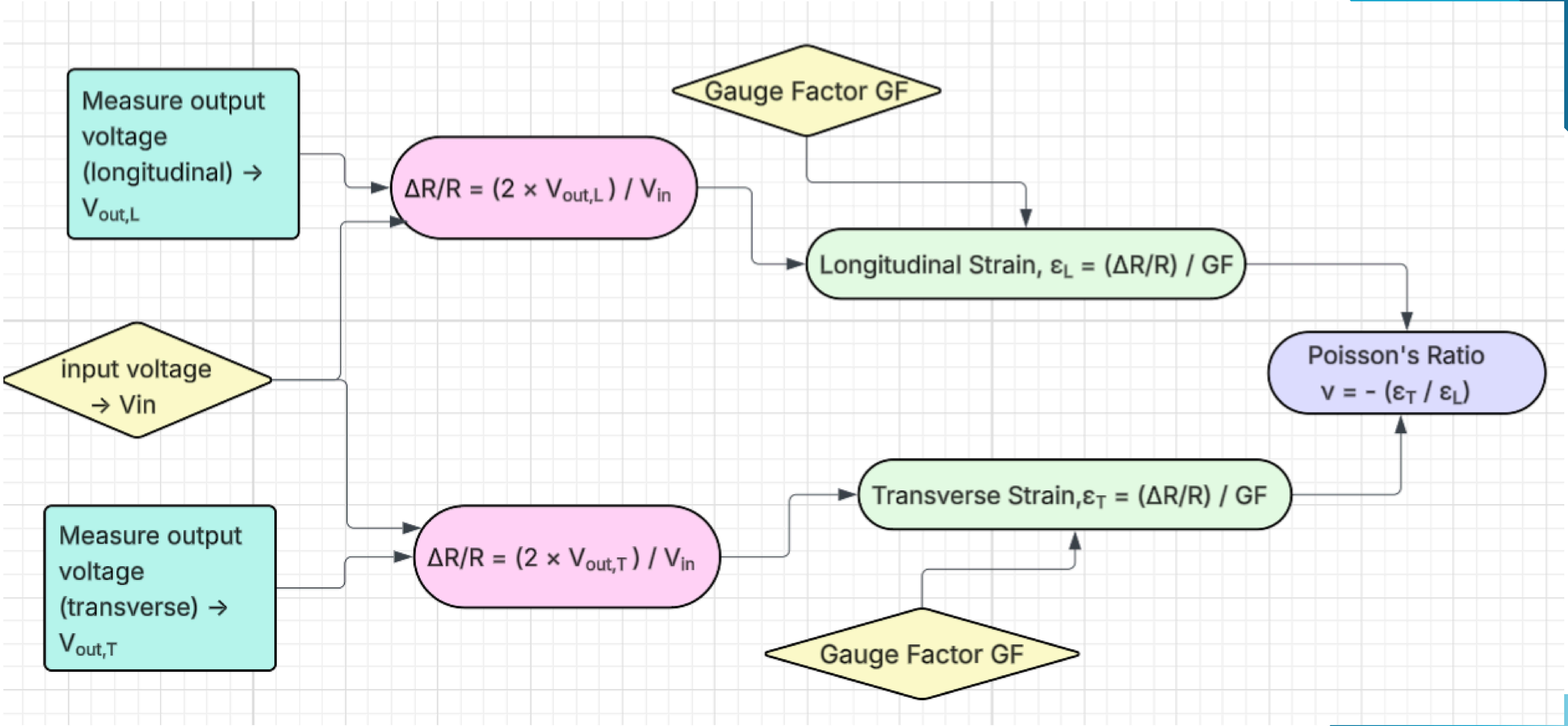


1. Basic Principle::

When a material is subjected to an external load, it undergoes deformation. This deformation is quantified as strain (ε), defined as:

$$\varepsilon = \frac{\Delta L}{L}$$

Within the elastic limit, stress and strain are related by Young's Modulus (E):

$$E = \frac{\sigma}{\varepsilon}$$

2. Strain Measurement Using Strain Gauges::

A strain gauge (350 Ω) works on the principle that its electrical resistance changes when stretched or compressed.

$$\frac{\Delta R}{R} = GF \cdot \varepsilon$$

Since the change in resistance is extremely small, it cannot be measured directly—hence the use of a Wheatstone bridge.

3. Half-Bridge Wheatstone Circuit::

In setup:

- Two arms contain strain gauges
- Two arms contain fixed resistors (350 Ω)
- This forms a half-bridge configuration, which improves:
- Sensitivity
- Temperature compensation

When load is applied:

- One gauge → tension (R increases)
- One gauge → compression (R decreases)
- This creates a voltage imbalance (Vout).

4. Bridge Output Relation (Key Step)::

For a half-bridge:

$$\frac{V_{out}}{V_{in}} = \frac{GF \cdot \varepsilon}{2}$$

Rearranging to get strain:

$$\varepsilon = \frac{2V_{out}}{GF \cdot V_{in}}$$

5.Signal Amplification and Data Acquisition::

The output voltage from the Wheatstone bridge is very small (in microvolts). Therefore, an HX711 amplifier is used to amplify the signal and convert it into digital form. The ESP32 microcontroller reads this digital value as a raw output.

The raw value obtained from the HX711 is proportional to the output voltage:

$$V_{out} = k \times \text{Raw Value}$$

where k is the calibration factor.

6.Bending Theory of Beam::

For a beam subjected to bending, the strain is related to the applied load by:

$$\varepsilon = \frac{6FL}{Ebh^2}$$

7.Final Expression for Young's Modulus::

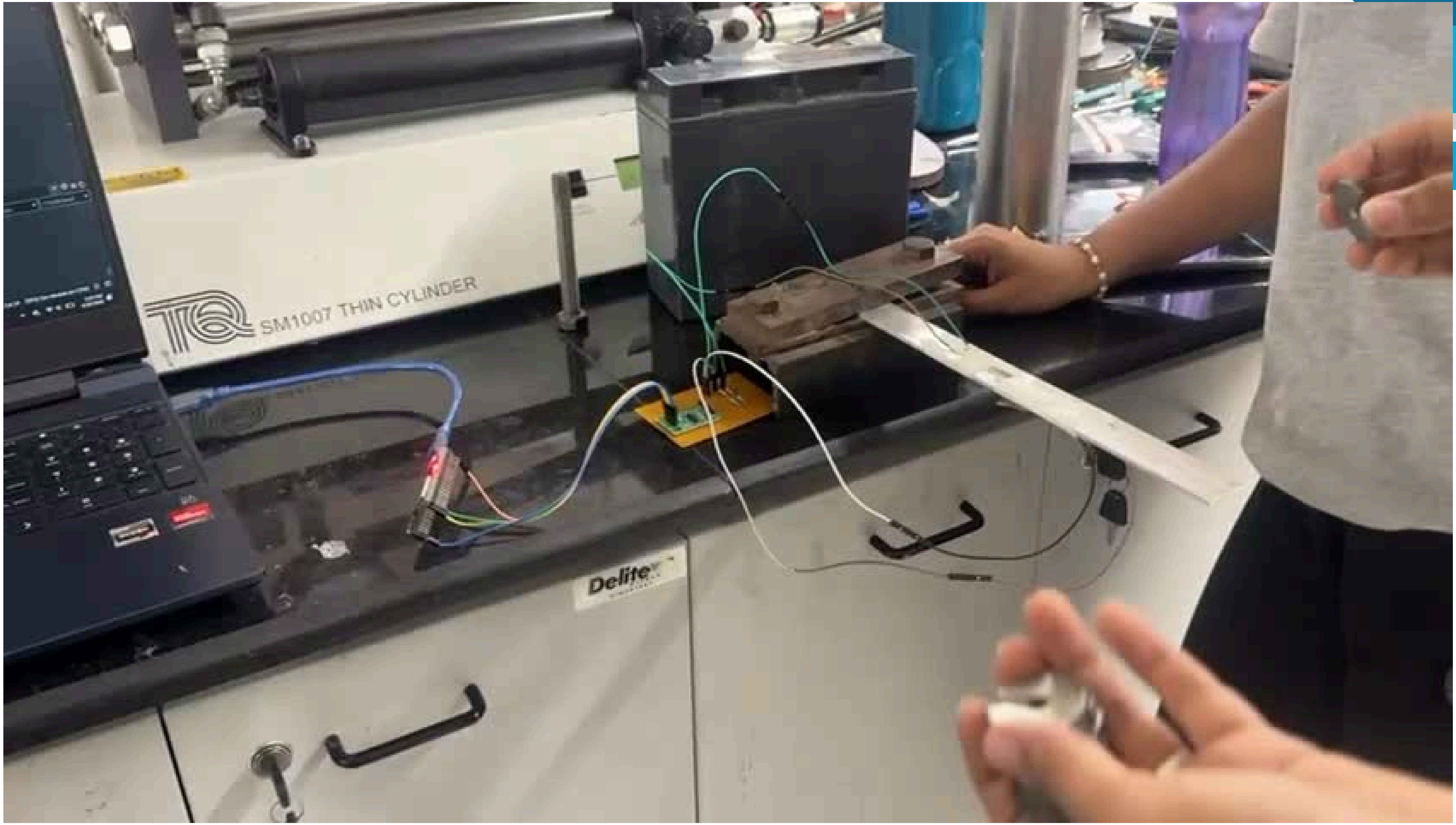
By substituting the strain value obtained from the Wheatstone bridge:

$$E = \frac{3FL \cdot GF \cdot V_{in}}{bh^2 \cdot V_{out}}$$

WORKING PRINCIPLE ::

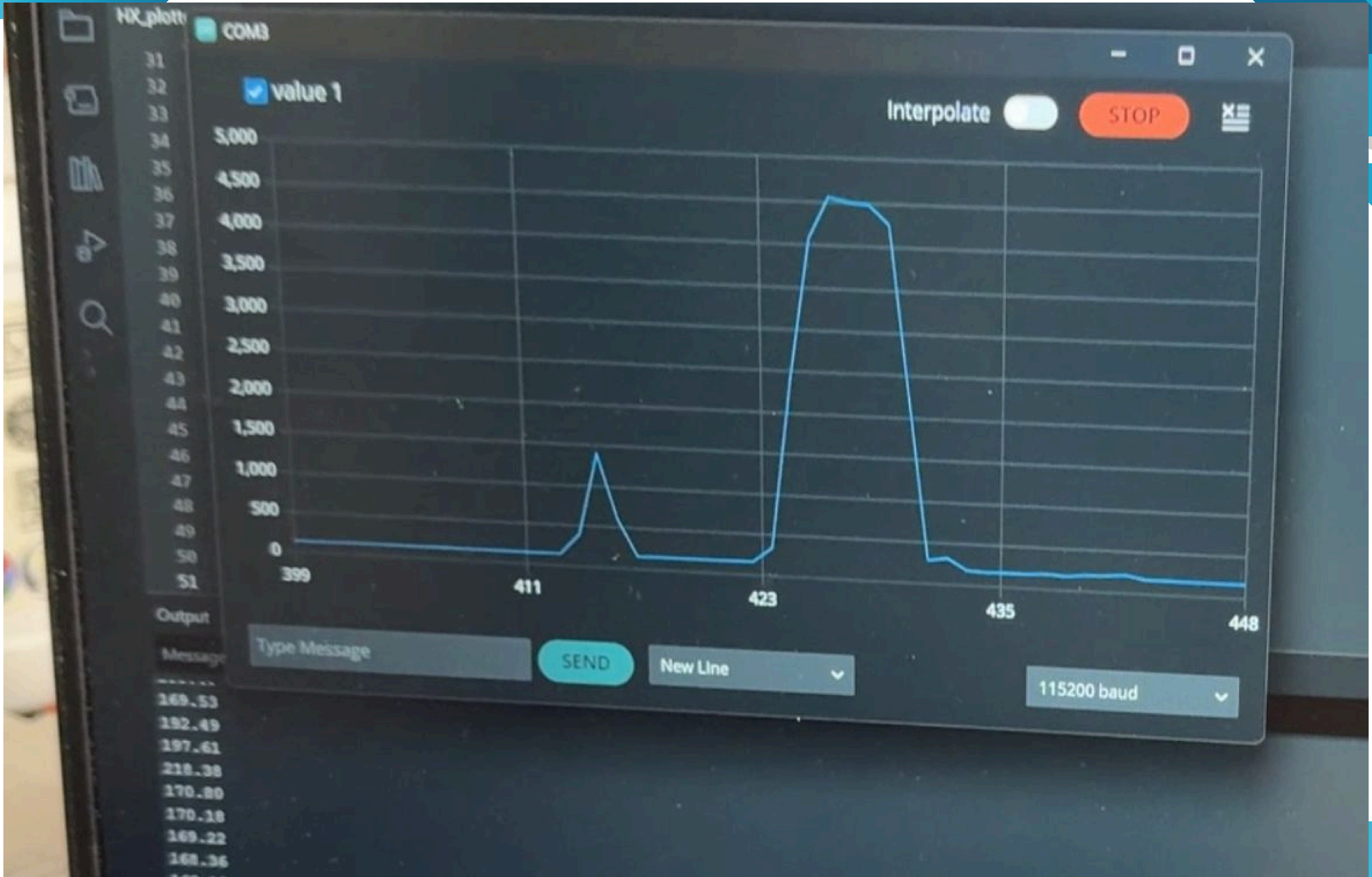
The complete working of the system can be summarized as follows:

- Load is applied to the beam
- The beam undergoes deformation, producing strain
- Strain gauges convert deformation into resistance change
- Wheatstone bridge converts resistance change into voltage
- HX711 amplifies the voltage signal
- ESP32 reads the digital (raw) value
- Raw value is converted into voltage
- Voltage is used to calculate strain
- Strain is used to determine Young's Modulus


SM1007 THIN CYLINDER
Delite

Using 10 g weights

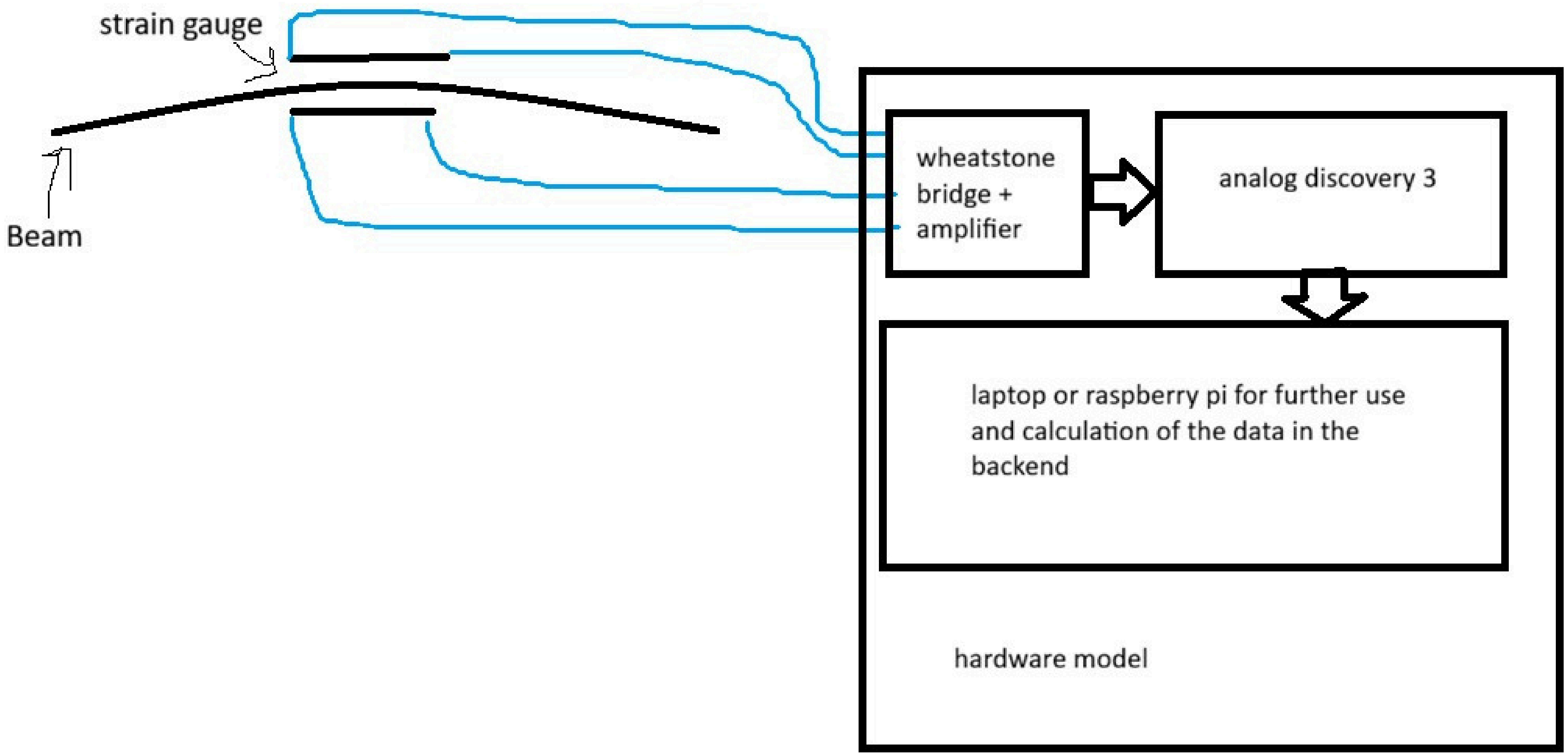
Using 200 g weights


COM3
value 1
Interpolate
STOP
5,000
4,500
4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
0
399
411
423
435
448
Output
Type Message
SEND
New Line
115200 baud
169.53
192.49
197.61
218.38
170.80
170.18
169.22
168.36

WEEK-BY-WEEK PROGRESS

Week 1:

Conducted detailed background research on the project. Studied the working principle of strain gauges, the concept of Wheatstone bridges, and methods of measuring small resistance changes. Identified the required components and planned the overall system architecture.



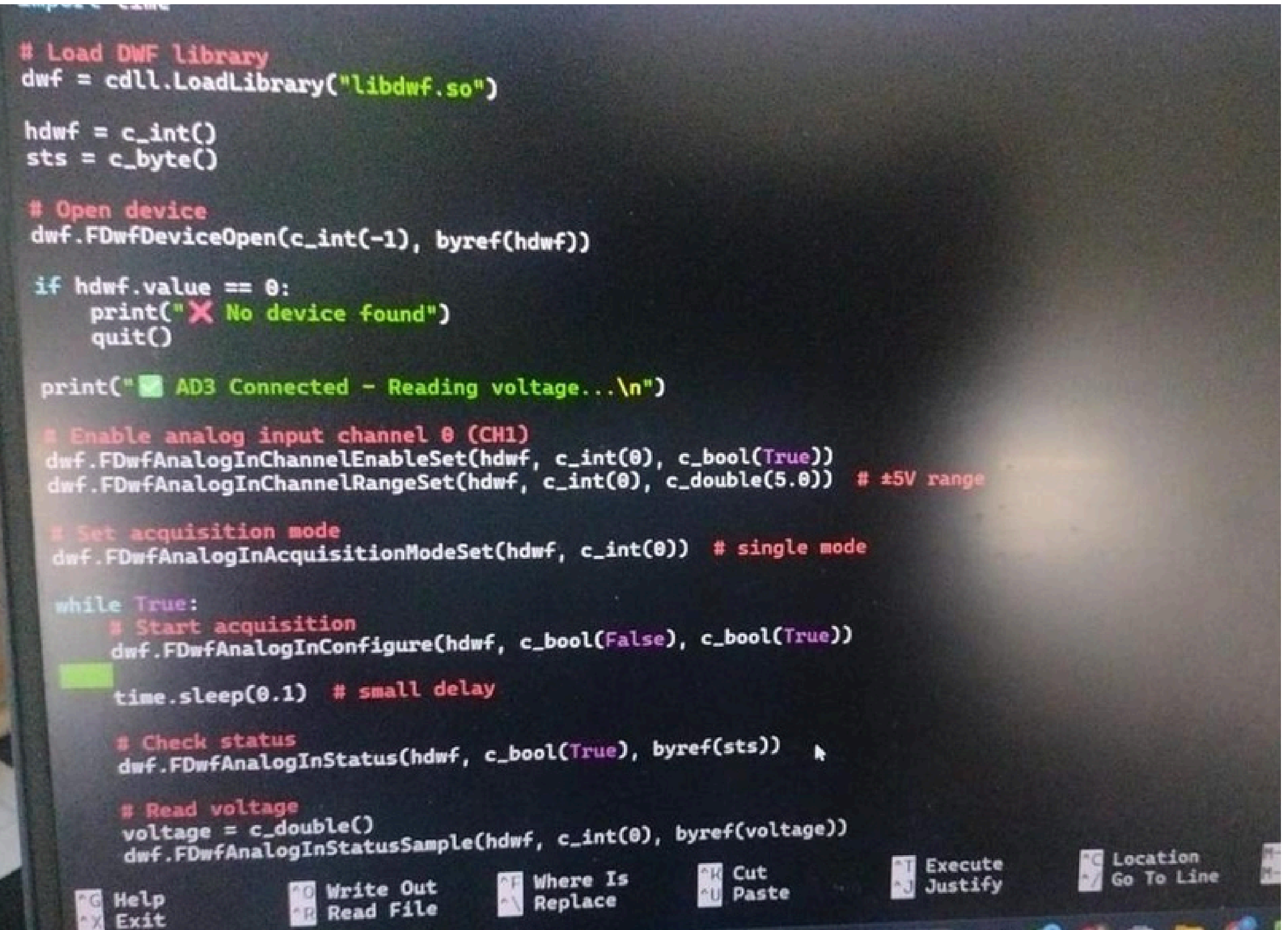
WEEK-BY-WEEK PROGRESS

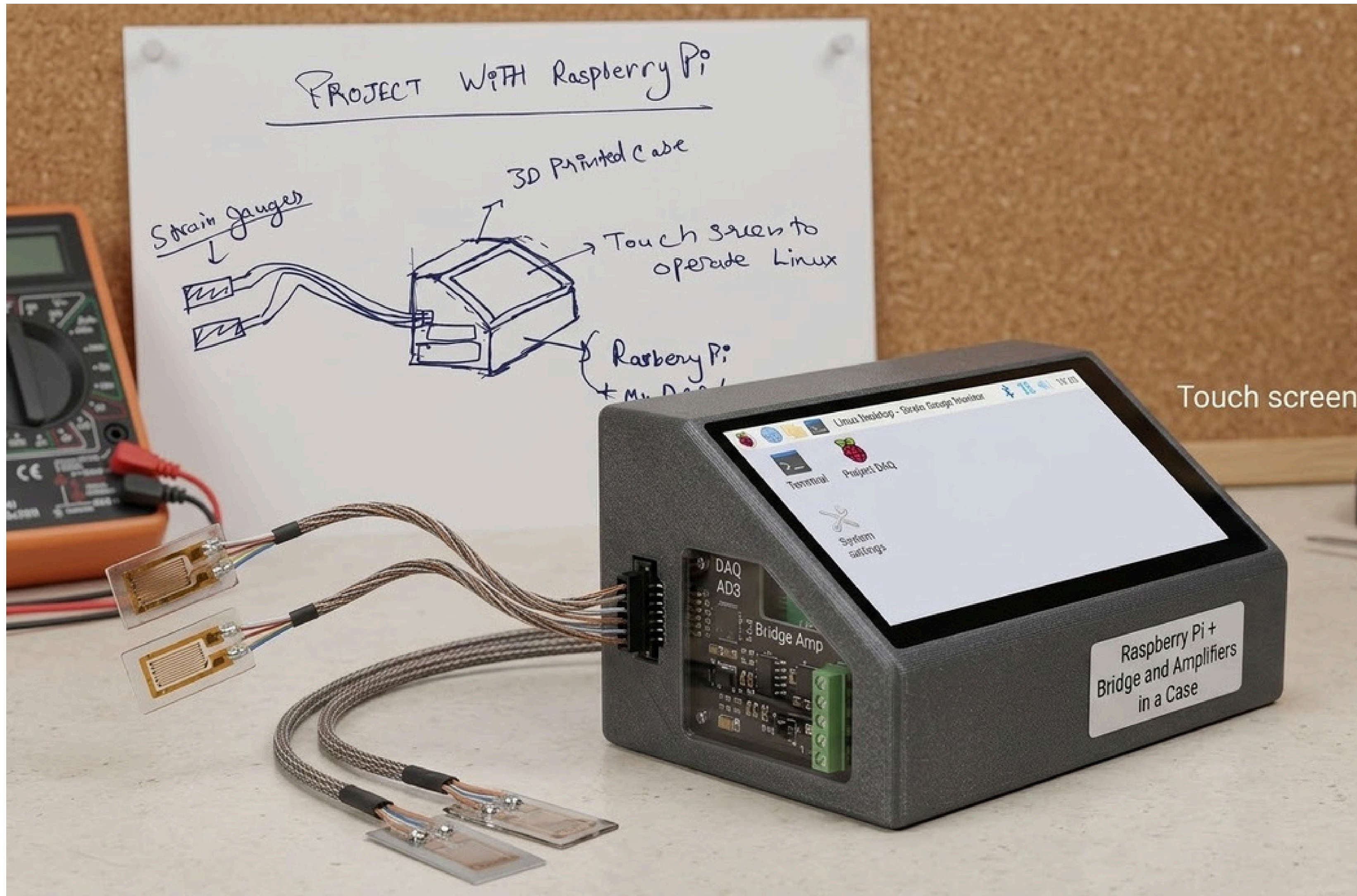
Week 2:

Explored the Y3 module in depth, including its specifications, pin configuration, and expected output behavior. Attempted initial interfacing and understood its role in signal acquisition.

Week 3:

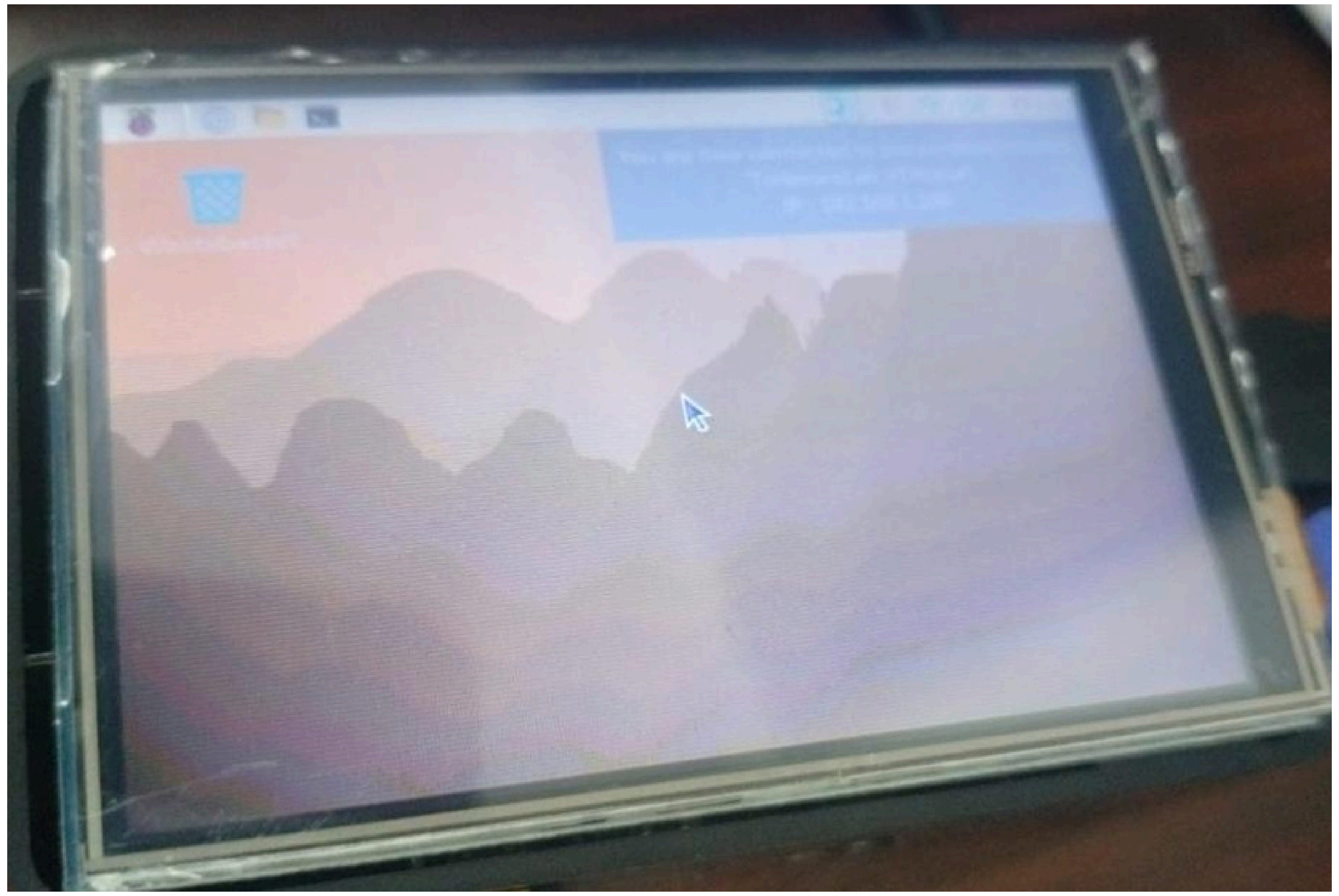
Studied the AD3 module and successfully interfaced it with the Raspberry Pi. Tested basic data acquisition and ensured communication between hardware and software was functioning correctly.





WEEK-BY-WEEK PROGRESS

Week 4 : Developed a simple graphical/user interface on the Raspberry Pi with a display screen. The UI was designed to show real-time readings and improve user interaction with the system.



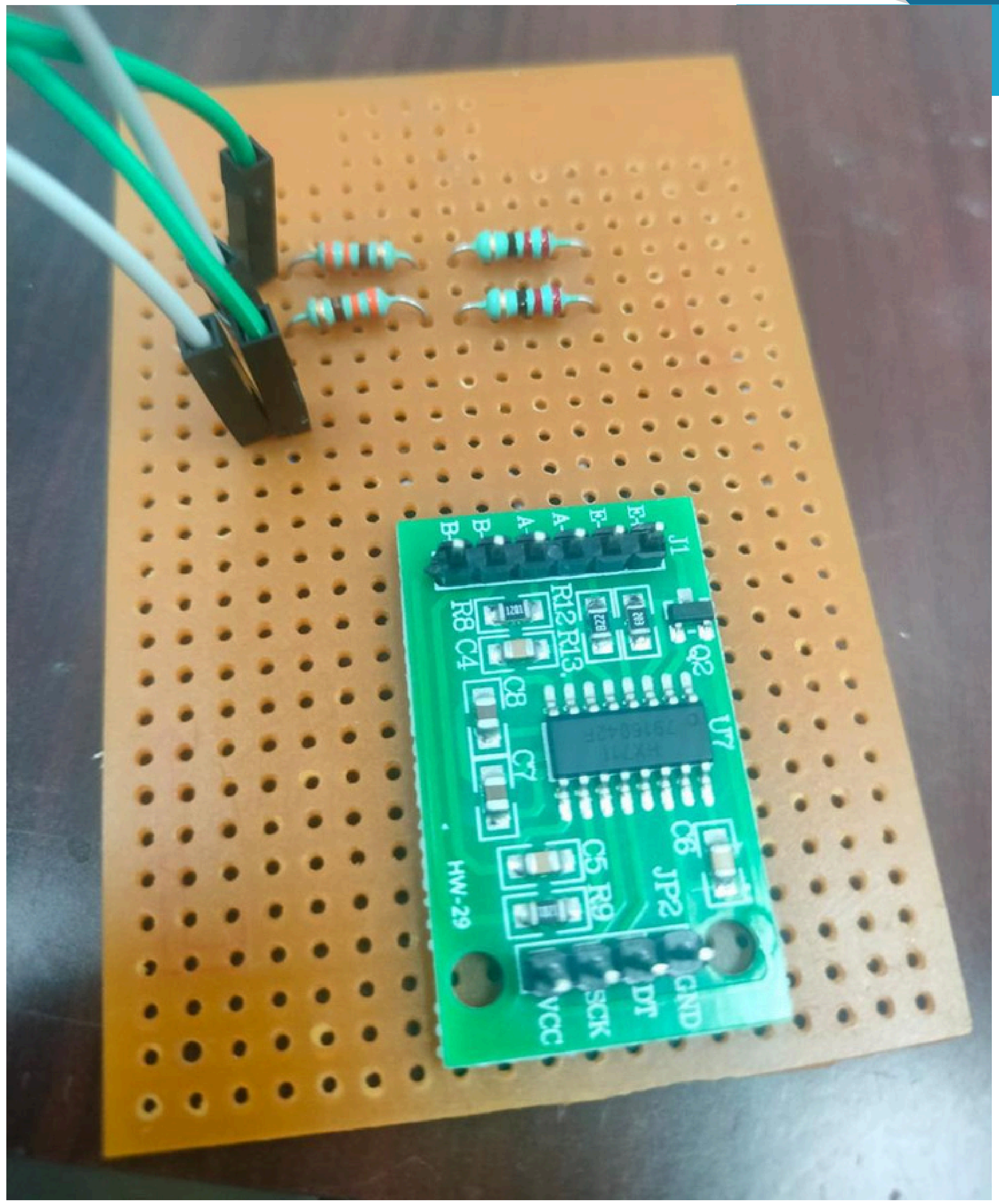
Week 5: Conducted initial experimental testing by applying load to the setup. Observed system response and attempted to correlate applied force with output readings, though results were inconsistent.

Week 6: Faced major issue where no meaningful data was obtained. Performed systematic debugging of the entire setup, including circuit connections, wiring, and code. After thorough analysis, identified that the newly provided Y3 modules and some lab components were faulty and not producing output.

WEEK-BY-WEEK PROGRESS

Week 7 : Revised the approach by designing and implementing a Wheatstone bridge circuit using a 350 Ω strain gauge. Studied and integrated the HX711 to amplify the small voltage output from the bridge and convert it into digital signals suitable for processing.

Week 8: Resolved wiring and connection issues, ensured proper grounding and stable connections, and calibrated the system. Successfully obtained consistent and reliable output readings corresponding to applied load, completing the experimental setup.



OBSERVATIONS:

1. When no load was applied to the 350 Ω strain gauge, the half-bridge was nearly balanced and the output of the HX711 remained close to a constant baseline value (offset).
2. On applying load, the resistance of the active strain gauge changed slightly, causing an imbalance in the bridge and generating a small differential voltage at the output.
3. The output reading from the HX711 changed consistently with the applied load, showing an approximately linear relationship between load (strain) and output signal.
4. Small fluctuations were observed in the readings due to:
   - Electrical noise
   - Vibrations or inconsistent loading
   - Temperature variations
5. The HX711 successfully amplified the low-level (millivolt) signal from the bridge, making it possible to obtain stable digital readings.
6. Slight offset was present in the initial readings, which could be minimized by calibration or balancing of the bridge.
7. Repeated application and removal of load produced nearly similar readings, indicating reasonable repeatability, though minor deviations were observed.
8. The system responded quickly to changes in load, indicating good dynamic response for small strain variations.

ISSUES FACED:

- Frequent failure of electrical components:
- Some components (wires, connections, or modules like the HX711) showed unstable behavior or stopped working, possibly due to loose connections or improper handling.
- Strain gauge was very delicate:
- The 350 Ω strain gauge was extremely fragile and prone to damage during handling, making installation and usage difficult.
- Difficulty in mounting (pasting) the strain gauge:
- Properly attaching the strain gauge to the surface was challenging. Poor bonding affected strain transfer and led to inaccurate readings.
- Noise due to wires and connections:
- Long or loose wires introduced electrical noise, causing fluctuations and unstable output readings

CONCLUSION

This project focused on measuring Young's modulus and Poisson's ratio using strain gauges integrated with a Wheatstone bridge, HX711 amplifier, and ESP32 microcontroller. The system was able to detect and record variations in strain through corresponding changes in output voltage. However, the determination of Young's Modulus and Poisson's Ratio could not be completed due to calibration limitations. In particular, the gauge factor, which is essential for converting electrical signals into accurate strain values, was not experimentally determined. As a result, only relative strain measurements were obtained. Future work will involve proper calibration, experimental evaluation of the gauge factor, and refinement of the setup to enable complete material property analysis.

THANK YOU